Q09: Data Composition Shapes What a Model Knows

Paradise (nuc13)

October 2026

The question

- "How does the composition of training data shape what a model knows and can do?"
- Collection/curation, dedup, filtering, mixture weights, curriculum, quality vs. scale
- Answer preview: **at fixed compute, composition is among the cheapest levers — and sometimes rivals modest scale increases**

What is in the corpus?

- Common Crawl snapshots dominate by volume — raw HTML, uneven quality
- Curated slices: code (The Stack: 3.1 TB, 30 languages; Kocetkov et al. 2022), Wikipedia, books, papers
- Heterogeneous sources broadly beneficial; single-source curricula worse (Longpre et al. 2023)
- A “token” is not fungible: learning signal per FLOP varies by source

Deduplication: exact + MinHash

- Exact: hash documents/lines — cheap, misses boilerplate and paraphrase
- Fuzzy (MinHash): shingle sets; Pr[signature match] = Jaccard(A, B); LSH banding makes it subquadratic
- Why dedup: fewer wasted steps, $\sim$10$\times$ less verbatim regurgitation, honest eval (Lee et al. 2022)
- But: global dedup of 96 CC snapshots cut 20T$\rightarrow$4T tokens and trained *worse* than per-snapshot dedup (Penedo et al. 2024)

Quality filtering: three families

1. **Heuristics** (C4-style): length, punctuation, stopword rules — cheap, blunt
2. **Perplexity** (CCNet; Wenzek et al. 2020): keep text a Wikipedia-trained LM finds likely — domain targeting in disguise
3. **Classifier** (FineWeb-Edu; Penedo et al. 2024): LLM-annotated "educational value"; 15T $\rightarrow$ 1.3T tokens

Result at fixed 350B-token budget: MMLU 33%$\rightarrow$37%, ARC 46%$\rightarrow$57%
Caveat: filters trade benchmarks vs. coverage/toxicity; no universal threshold (Longpre et al. 2023)

Mixture weights and repetition

- DoReMi (Xie et al. 2023): group-DRO proxy model finds domain weights; +6.5 pts few-shot, 2.6× fewer steps at 8B
- Code in the mix: up to 50% Python $\Rightarrow \sim 2\times$ effective tokens *on natural-language tasks* (Muennighoff et al. 2023)
- Repetition: up to $\sim$4 epochs $\approx$ fresh data; then returns decay to zero (400+ runs, $\leq$9B params)
- Data-constrained optimum: smaller model, more epochs — not Chinchilla's equal scaling (Hoffmann et al. 2022)

Annealing and curriculum

- Annealing: diverse mix early $\rightarrow$ upsampled high-quality data at the end (LR decay)
- MiniCPM (Hu et al. 2024): math+code share raised to $\sim$70% in the annealing phase
- Data age: train/eval temporal gap degrades performance; finetuning does not repair it (Longpre et al. 2023)
- RHO-LOSS (Mindermann et al. 2022): ordering by reducible holdout loss accelerates training on noisy web data
- Honest read: coarse mixture schedules (annealing) $>$ fine-grained curricula at scale

Tokenizer–mixture interaction

- Tokenizer is trained on a sample of the mixture; fertility differs by domain
- Code / non-English text: more subwords per word $\Rightarrow$ fixed token budget holds less content
- Domain weights in tokens $\neq$ weights in bytes/characters
- Missing vocabulary biases the *effective* mixture away from the declared one
- Magnitude vs. mixture weights: unquantified at scale — open question

Why quality rivals scale

- Fresh-data stock could be fully utilized $\sim$2026–2032 (Villalobos et al. 2024)
- Repeated + code tokens keep working, with quantified diminishing returns (Muennighoff et al. 2023)
- Classifier filtering reorders benchmark rankings at fixed compute (Penedo et al. 2024)
- Optimized mixtures $\approx$ nontrivial compute increases (Xie et al. 2023)
- But the compute-optimal frontier still binds (Hoffmann et al. 2022) — quality complements, not replaces, scale

Reading

- Penedo et al. 2024 (FineWeb / FineWeb-Edu) — curation ablations at scale
- Muennighoff et al. 2023 — data-constrained scaling, 4-epoch rule, code augmentation
- Lee et al. 2022 — dedup, memorization, efficiency
- Hoffmann et al. 2022 (Chinchilla) — the compute-optimal baseline
- Longpre et al. 2023 — controlled curation ablations (age, domain, quality, toxicity)